

PROGRESSIONS

Directions for questions 1 and 2: Select the correct alternative from the given choices.

Find the 12th and 77th terms in the series 10, 12, 14, 16,

☒ a) 32, 162 (Your answer - ✓)

☐ b) 172, 32

☐ c) 34, 162

☐ d) 34, 68

Directions for questions 1 and 2: Select the correct alternative from the given choices.

The first term of an arithmetic progression is -15 and its 46th term is 120. What are the values of the 23rd term and the 6th term?

☐ a) 54, 0

☒ b) 51, 0 (Your answer - ✓)

☐ c) 51, 3

☐ d) 54, 3

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

Nine times the ninth term of an arithmetic progression is equal to ten times the tenth term of the same series. Find the nineteenth term of the series.

Your Answer: 0 ✓

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

Find the arithmetic mean of the first forty four multiples of forty four.

Solution:

The given numbers are

$$1(44), 2(44), 3(44), \dots, 44(44)$$

The mean of 1, 2,, 44 is $\frac{1+44}{2} = 22.5$

The mean of the given numbers is 44 (22.5)
 $= 11(90) = 990$ Ans : (990)

Ans : (990)

Directions for question 5: Select the correct alternative from the given choices.

Find the geometric mean of the first sixteen powers of sixteen.

- ☐ a) 2^{38}
- ☐ b) 2^{34}
- ☐ c) 2^{17}
- ☐ d) 2^{36}

Solution:

The given sequence is the GP $16, 16^2, \dots, 16^{16}$.

The GM is the 16th root of the product

$$\text{i.e. } 16^{\frac{1+2+\dots+16}{16}} = 16^{\frac{16(17)}{2(16)}} = 16^{8.5} = 2^{34}$$

Choice (B)

Directions for question 6: Type in your answer in the input box provided below the question.

What is the sum to 25 terms of an arithmetic progression whose first term is 5 and common difference is $\frac{7}{4}$?

Solution:

The first term of the AP is 5 and the common difference is $\frac{7}{4}$.

$$\therefore t_{25} = 5 + 24\left(\frac{7}{4}\right) = 47$$

The average of the first 25 terms = $(5 + 47)/2$
= 26

The sum of the first 25 terms = $25(26) = 650$

Ans: 650

Directions for questions 7 and 8: Select the correct alternative from the given choices.

Find three terms of an arithmetic progression such that their sum is 30 and product is 910.

- ☐ a) 8, 10, 12
- ☐ b) 7, 10, 13
- ☐ c) 6, 10, 14
- ☐ d) 5, 13, 14

Directions for questions 7 and 8: Select the correct alternative from the given choices.

Find three terms of an arithmetic progression such that their sum is 30 and product is 910.

- ☐ a) 8, 10, 12
- ☐ b) 7, 10, 13
- ☐ c) 6, 10, 14
- ☐ d) 5, 13, 14

Solution:

The sum of 3 terms of the AP is 30.
 $\therefore$ The average i.e. the second term is 10.
Also $(10 - x)10(10 + x) = 910 \Rightarrow x = \pm 3$.
The 3 terms could be 7, 10, 13 or 13, 10, 7.

Choice (B)

Directions for questions 7 and 8: Select the correct alternative from the given choices.

Three terms are in harmonic progression such that the sum of their reciprocals is equal to 36.

If the ratio of the first term to the last term is 2 : 1, find the numbers.

- ☐ a) $1/3, 1/12, 1/21$
- ☐ b) $1/4, 1/12, 1/20$
- ☐ c) $1/2, 1/12, 1/22$
- ☐ d) $1/8, 1/12, 1/16$

Solution:

Let the reciprocals of the first and third term be $2a$ and $4a$. (The ratio of the terms is 2 : 1).
Therefore, the second term (of the reciprocals) is $3a$. The sum of these reciprocals
 $= 4a + 3a + 2a = 36$. $\therefore a = 4$. The reciprocals are 8, 12 and 16.
The numbers are $1/8, 1/12, 1/16$

Choice (D)

Directions for question 9: Type in your answer in the input box provided below the question.

The eighth term of a G.P. is 27 times its fifth term. If the third term of the G.P. is 54, find its first term.

Solution:

If the first term and the common ratio of the GP are a, r respectively, the fifth and eighth terms are ar^4 and ar^7 respectively.
 $ar^7 = 27 ar^4$. Also $ar^2 = 54$
 $\therefore r = 3$ and $a = 6$

Ans : (6)

Directions for questions 10 to 13: Select the correct alternative from the given choices.

What is the sum of all the multiples of nine lying between 300 and 800?

- ☐ a) 29646
- ☐ b) 49505
- ☐ c) 49595
- ☐ d) 30195

Solution:

$$\begin{aligned}\text{Let } S &= 306 + 315 + \dots + 792 \\ &= 9(34 + 35 + \dots + 88) \\ &= 9(55) \left(\frac{34 + 88}{2} \right) = 9(55) (61) = 30195\end{aligned}$$

Choice (D)

Solution:

$$\begin{aligned}\text{Let the 5 terms be } a - 2d, a - d, a, a + d, a + 2d. \\ \text{As the sum is 100, } a = 20. \text{ Also, } a^2 - 4d^2 = 144 \\ \Rightarrow 4d^2 = a^2 - 144 \Rightarrow 4d^2 = 400 - 144 = 256 \Rightarrow d =\end{aligned}$$

± 8
Choice (C)

Directions for questions 10 to 13: Select the correct alternative from the given choices.

Find the sum to five terms of a G.P. whose first term is 4 and common ratio is $\frac{3}{2}$.

- ☐ a) $183/32$
- ☐ b) $223/16$
- ☐ c) $211/4$
- ☐ d) $169/4$

Solution:

The given GP is 4, 6, 9, 13.5, 20.25. The sum of 5 terms = $52\frac{3}{4} = \frac{211}{4}$
Choice (C)

Directions for questions 10 to 13: Select the correct alternative from the given choices.

Find the 12th term of a geometric progression whose 5th term is 144 and the 8th term is $-\frac{128}{3}$.

- ☐ a) $-2048/243$
- ☐ b) $1024/729$
- ☐ c) $-2048/729$
- ☐ d) $512/81$

Solution:

Let the first term and the common ratio of the GP be a, r respectively
 $ar^4 = 144$ (1)

$$ar^7 = \frac{-128}{3} \text{(2)}$$

$$(2) \div (1) \Rightarrow r^3 = \frac{-128}{3} \cdot \frac{1}{144} = \frac{-8}{27} \Rightarrow r = \frac{-2}{3}$$

$\therefore$ the 12th term = $(ar^7)(r^4)$

$$= \left(\frac{-128}{3}\right)\left(\frac{16}{81}\right) = \frac{-2048}{243}$$

Choice (A)

Directions for question 14: Type in your answer in the input box provided below the question.

Find the first term of a G.P. whose sum to infinity is $5\frac{1}{2}$ and second term is $11/8$.

Solution:

Let the first term and the common ratio of the GP be a, r respectively.

$$ar = \frac{11}{8} \quad \text{--- (1) and}$$

$$\frac{a}{1-r} = \frac{11}{2} \quad \text{--- (2)}$$

$$(1) \div (2) \Rightarrow r(1-r) = \frac{1}{4} \Rightarrow 4r^2 - 4r + 1 = 0$$

$$\Rightarrow (2r - 1)^2 = 0 \Rightarrow r = \frac{1}{2}$$

$$\therefore a = \frac{11}{8}(2) = \frac{11}{4}$$

Ans : (2.75)

Directions for question 15: Select the correct alternative from the given choices.

Find the 12th term of a harmonic progression whose sixth term is $1/18$ and tenth term is $1/24$.

- ☐ a) $1/30$
- ☐ b) $1/27$
- ☐ c) $2/51$
- ☐ d) $2/53$

Solution:

Let the HP be $\frac{1}{a}, \frac{1}{a+d}, \dots$. We need the value of $\frac{1}{a+11d}$.

$$\frac{1}{a+5d} = \frac{1}{18} \text{ and } \frac{1}{a+9d} = \frac{1}{24}$$

$$\therefore a + 5d = 18 \text{ and } a + 9d = 24, \text{ i.e. } d = 1.5, a = 10.5$$

$$a + 11d = 10.5 + 11(1.5) = 10.5 + 16.5 = 27.$$

The 12th term of the HP is $1/27$.

Choice (B)